BLOCKCHAIN

DEFINITION:

Blockchain is a decentralized digital ledger that securely stores records across a network of computers in a way that is transparent, immutable, and resistant to tampering.

or

Blockchain is a digital ledger that records and verifies transactions across a network of computers. It's a decentralized system that doesn't rely on a central authority to confirm transactions.

- They are best known for their crucial role in cryptocurrency systems, maintaining a secure and decentralized record of transactions.
- Blockchains can be used to make data in any industry immutable—meaning it cannot be altered.
- blockchains store data in blocks linked together via cryptography.
- A block is continuously growing list of ordered records.
- Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data

- A cryptographic hash of the previous block is a unique fixed-length identifier generated using a cryptographic hash function (like SHA-256) from the data of the previous block in a blockchain. It ensures data integrity, prevents tampering, and links blocks together securely.

- Consensus of the network is the process by which all nodes in a blockchain agree on the validity of transactions and the state of the ledger. Common consensus mechanisms include Proof of Work (PoW) and Proof of Stake (PoS), ensuring security and decentralization.

- Proof of Work (PoW) is a consensus mechanism in blockchain where miners solve complex mathematical puzzles to validate transactions and add new blocks. It ensures security and prevents fraud but requires high computational power and energy. Bitcoin uses PoW.

- Proof of Stake (PoS) is a blockchain consensus mechanism where validators are chosen to create new blocks based on the amount of cryptocurrency they "stake" as collateral. It is energy-efficient compared to Proof of Work (PoW) and enhances security while reducing computational costs.

How Does a Blockchain Work?

- **Transaction Initiation** – A user requests a transaction (e.g., sending cryptocurrency).
- **Transaction Verification** – Network nodes (computers) validate the transaction using consensus mechanisms like **PoW** or **PoS**.
- **Block Creation** – Verified transactions are grouped into a block.
- **Hashing and Linking** – Each block contains a unique cryptographic hash of the previous block, ensuring security.
- **Consensus and Addition** – The network agrees on the block's validity, and it is added to the blockchain.
- **Ledger Update** – The updated blockchain is distributed across all network nodes.

How Does a Blockchain Work?

- A blockchain is distributed, which means multiple copies are saved on many machines, and they must all match for it to be valid.

- The Bitcoin blockchain collects transaction information and enters it into a 4MB file called a block (different blockchains have different size blocks).

- Once the block is full, the block data is run through a cryptographic hash function, which creates a hexadecimal number called the block_header hash.

- The hash is then entered into the following block header and encrypted with the other information in that block's header, creating a chain of blocks, hence the name “blockchain.”

Transaction Process

Miners are participants in a blockchain network who use computational power to solve cryptographic puzzles in Proof of Work (PoW) systems. They validate transactions, add new blocks to the blockchain, and earn rewards (block rewards + transaction fees).

In Proof of Work (PoW), miners compete to find a valid hash that meets the difficulty target by adjusting a variable called the nonce (Number Used Once).

How it works:

- Each miner works on their own block, trying different nonce values.
- The miner repeatedly hashes the block data (including the nonce) until the result meets the required difficulty target (i.e., a hash with a certain number of leading zeros).
- Once a valid hash is found, the miner broadcasts the block to the network.
- Other nodes verify the solution, and if valid, the block is added to the blockchain.
- The successful miner earns a block reward and transaction fees.

Transaction Process

The difficulty target in blockchain (especially in Proof of Work) is a predefined value that determines how hard it is to find a valid block hash. It ensures that new blocks are added at a consistent rate.

How it Works:

- **Miners must find a hash value lower than or equal to the difficulty target.**

- **The difficulty adjusts periodically (e.g., every 2016 blocks in Bitcoin) based on network conditions to maintain a steady block time (e.g., 10 minutes for Bitcoin).**

- **Higher difficulty means more computational power is required to mine a block.**

- **This mechanism prevents blocks from being mined too quickly and maintains network stability.**

Applications of Blockchain

- **Cryptocurrency** – Powers digital currencies like **Bitcoin** and **Ethereum** for secure transactions.
- **Smart Contracts** – Self-executing contracts on platforms like **Ethereum** automate agreements.
- **Supply Chain Management** – Ensures transparency and tracking in logistics (e.g., Walmart, IBM Food Trust).
- **Healthcare** – Secures patient records, ensuring data integrity and privacy.
- **Banking & Finance** – Enables **fast, secure cross-border payments** (e.g., Ripple, DeFi).
- **Voting Systems** – Provides **tamper-proof digital voting** for fair elections.
- **Real Estate** – Facilitates **secure property transactions** via blockchain-based ledgers.
- **Identity Management** – Prevents identity fraud with decentralized digital IDs.

Pros	Cons
Improved accuracy by removing human involvement in verification	Significant technology cost associated with some blockchains
Cost reductions by eliminating third-party verification	Low number of transactions per second
Decentralization makes it harder to tamper with	History of use in illicit activities, such as on the dark web
Transactions are secure, private, and efficient	Regulation varies by jurisdiction and remains uncertain
Transparent technology	Data storage limitations